

THE ZERO PARADOX

ZP-L: Incomputability Convergence

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All theorems §I–§VII proved sorry-free in Lean 4. Axiom footprint: [propext, Classical.choice, Quot.sound] throughout this document’s own Lean file (Gentzen.lean). Across the ZP layers the footprints are not uniform — ZP-K Section IV tabulates them.

ZP-L establishes four results connecting the formal axioms of the ZP framework to standard results in ordinal theory and computability. First, the axiom footprints of the layers tabulated in Section I are surveyed, and they are not uniform; ZP-K Section IV tabulates the measured footprints. Second, Rogers’ fixed-point theorem for a total computable transformation (Mathlib fixed_point; inter-derivable with Kleene’s second recursion theorem, fixed_point₂) is formalized as a wrapper, formalizing the computational fixed-point structure. Third, the ordinal ε_0 is fully characterized as the first fixed point of $\alpha \mapsto \omega^\alpha$ and the limit of the tower $\omega, \omega^\omega, \omega^{\omega^\omega}, \dots$. Fourth, ordinals below ε_0 encode into $\mathbb{Z}_2$ via their Cantor normal form, and as the tower stages approach ε_0 , their 2-adic encodings converge to $0 = \perp$.

The central result (§VII) is the canonical snap map: $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$ simultaneously satisfies all five conditions — monotone, tower-aligned, fixed-point-respecting, snapping at ε_0 , and ε_0 minimal. All conditions are verified without free hypotheses for this witness. Proved sorry-free, axiom footprint [propext, Classical.choice, Quot.sound] throughout ZeroParadox/Ordinal/Gentzen.lean, measured.

Section I: Axiom Footprint Convergence

Non-constructibility appears across the layers listed below. The axiom footprints are not uniform: ZP-K Section IV tabulates the measured footprints, and the ZPJ/K row below names a witness on each side — bot_self_mem, which measures no axioms, and botCode, which carries them.

Layer	Formal Language	Expression of non-constructibility
ZPB	Topology	C3: no continuous path $\perp \rightarrow x \neq \perp$
ZPC	Information Theory	L-INF: infinite surprisal at $\perp$
ZPJ/K	Set Theory + Computation	bot_self_mem (AFA); botCode (Kleene)
ZPI	Algorithmic IT	$K(S_n n)/ S_n \rightarrow 1$; K uncomputable

Remark: K is not computed in Lean

Kolmogorov complexity K is not computed in Lean in this framework. The AFA/Kleene route reaches the same fixed-point structure via a path whose Kleene step is a KleeneStructure requirement, without requiring K to be explicitly computed.

Remark: K is not computed in Lean

Axiom footprint evidence — the following ZP-K theorems all carry [propext, Classical.choice, Quot.sound]:

t_comp (T-COMP three-way equivalence; the computational face is an assumption, not a clause)

da1_paths_unified

isComputationalQuine_undecidable

infinite_quine_family

Gentzen.lean carries that footprint throughout, measured — and it is not inherited from the theorems above. Mathlib's ordinal theory carries it on its own: Ordinal.nfp and Ordinal.epsilon_zero_eq_nfp each measure [propext, Classical.choice, Quot.sound] with no computability module in scope. ZP-K Section IV tabulates the measured footprints for the computability side.

Section II: Rogers Fixed-Point Stability

Rogers' fixed-point theorem (Mathlib fixed_point; inter-derivable with Kleene's second recursion theorem, fixed_point₂) states that any total computable transformation of a code has a behavioral fixed point: a code c such that running $f(c)$ and running c produce the same partial function. For any computable transformation, at least one fixed-point code exists.

Theorem: roger_fixed_point_stability (Gentzen.lean §II)

For any computable $f : \text{Code} \rightarrow \text{Code}$,

$\exists c : \text{Code}, \text{eval } (f\ c) = \text{eval } c$

Proof: wrapper around roger_fixed_point_exists.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Note: this is an existential result. The specific code c is not constructively produced. ZP-K Section IV tabulates the measured footprints.

Section III: The Ordinal ϵ_0

The ordinal ϵ_0 is the smallest fixed point of the map $\alpha \mapsto \omega^\alpha$. It is the supremum of the tower $0, 1, \omega, \omega^\omega, \omega^{\omega^\omega}, \dots$ — each stage is strictly below ϵ_0 , and ϵ_0 is not reached by any finite iteration. This section is entirely within Lean scope via Mathlib's ordinal machinery.

Definitions (Gentzen.lean §III)

epsilonZero : Ordinal := Ordinal.epsilon 0 (= nfp ($\omega^\cdot$) 0 = veblen 1 0)

fundamentalSeq : $\mathbb{N} \rightarrow \text{Ordinal} := \text{fun } n \mapsto (\alpha \mapsto \omega^\alpha)^\wedge[n] 0$

Definitions (Gentzen.lean §III)

Explicit stages:

$$\text{fundamentalSeq } 0 = 0$$

$$\text{fundamentalSeq } 1 = 1$$

$$\text{fundamentalSeq } 2 = \omega$$

$$\text{fundamentalSeq } 3 = \omega^\omega$$

$$\text{fundamentalSeq } (n+1) = \omega^{(\text{fundamentalSeq } n)}$$

Theorem: epsilonZero_fixedPoint

$$\omega^{\varepsilon_0} = \varepsilon_0$$

Proof: Ordinal.omega0_opow_epsilon 0 (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilonZero_eq_nfp

$$\varepsilon_0 = \text{nfp } (\omega^{\cdot}) 0$$

Proof: Ordinal.epsilon_zero_eq_nfp (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilonZero_eq_iSup

$$\varepsilon_0 = \bigsqcup n : \mathbb{N}, \text{fundamentalSeq } n$$

Proof: iSup_iterate_eq_nfp (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilonZero_tower_lt

$$\forall n : \mathbb{N}, \text{fundamentalSeq } n < \varepsilon_0$$

Every finite stage of the tower is strictly below ε_0 .

Proof: Ordinal.iterate_omega0_opow_lt_epsilon_zero (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilonZero_le_fixedPoint

$$\forall b : \text{Ordinal}, \omega^b = b \rightarrow \varepsilon_0 \leq b$$

Theorem: epsilonZero_le_fixedPoint

ε_0 is the least fixed point of $\omega^\cdot$ above 0.

Proof: Ordinal.epsilon_zero_le_of_omega0_opow_le (Mathlib).

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: fundamentalSeq_strictMono

$\forall n : \mathbb{N}, \text{fundamentalSeq } n < \text{fundamentalSeq } (n + 1)$

The tower is strictly monotone.

Proof: by induction; successor step uses isNormal_opow.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Remark R-L.1: Proof-Theoretic Alignment

Two results pin ε_0 as the proof-theoretic ordinal of Peano Arithmetic — the limit of how far PA's proofs of transfinite induction reach — and they run in opposite directions. Gentzen reports, crediting Hilbert-Bernays, that transfinite induction up to any ordinal strictly below ε_0 is provable in PA. And transfinite induction up to ε_0 itself is not provable in PA, which Gentzen proved directly in 1943. Neither half alone locates the boundary: a ceiling with no witnesses beneath it pins nothing, and witnesses with no ceiling pin nothing either. A third result stands behind them — transfinite induction up to ε_0 suffices to prove PA consistent (1936) — and it is also the indirect route to the unprovability half, via Gödel's theorem together with that 1936 result. This is not claimed or proved here.

ZP-L derives ε_0 as the snap threshold from ordinal fixed-point structure, independently of proof theory. Both derivations locate the same boundary: the ordinal where ω -tower self-iteration becomes self-limiting. No claim is made that this alignment is more than a structural observation.

Remark R-L.2: Surreal Numbers

Every ordinal is a surreal number (Conway, 1976), so ε_0 lives in the surreal number field No. No satisfies the axioms of a real-closed field, and by Tarski's completeness theorem for the theory of real-closed fields, every first-order sentence in the language of ordered fields that holds in $\mathbb{R}$ also holds in No, and vice versa.

ZP-F (The Counterexamples) proves that the Binary Snap — the transition from $\perp$ to the first non-null ordinal threshold (ε_0 , established in §V–§VII above) — cannot occur in any linearly ordered field. This result applies directly to No considered as a linearly ordered field. Note the scope: what is derived is the transition's SHAPE and its impossibility in an ordered field.

The surreals therefore contain ε_0 as an ordinal while simultaneously satisfying the density condition that blocks the Binary Snap in their ordered field structure. Both structures coexist in No: ε_0 is present as an ordinal, and the field density that blocks the snap is present in the field structure. The two results apply to different structural aspects of No.

Remark R-L.3: Hyperreals and Łoś's Theorem

The hyperreals ${}^*\mathbb{R} = \mathbb{R}^{\mathbb{N}}/U$ (ultrafilter U on $\mathbb{N}$) satisfy a transfer result: by Łoś's theorem, every first-order sentence true in $\mathbb{R}$ holds in ${}^*\mathbb{R}$. Łoś's theorem applies to the full first-order theory, so any first-order property of $\mathbb{R}$ — including field density — transfers.

Field density transfers: between any two hyperreals there is another. The density condition that blocks the Binary Snap in $\mathbb{R}$ (proved in ZP-F) therefore holds in ${}^*\mathbb{R}$ as well.

The non-standard naturals ${}^*\mathbb{N}$ contain infinite elements, but every infinite $H \in {}^*\mathbb{N}$ has a predecessor $H-1$ in ${}^*\mathbb{N}$. ϵ_0 is a limit ordinal: it has no predecessor and is the supremum of the tower below it. ${}^*\mathbb{N}$ cannot host the Binary Snap not only because of field density, but because its infinite elements are successor-like rather than limit-like. The snap requires genuine limit ordinal structure.

The monad of 0 in ${}^*\mathbb{R}$ — the external set $\{\epsilon : |\epsilon| < 1/n \text{ for all standard } n\}$ — is external to the ultrapower. Between any two distinct elements of the monad there is another (their arithmetic mean in ${}^*\mathbb{R}$), so no discrete jump at 0 is possible in ${}^*\mathbb{R}$.

Field density is an internal property of the ordered field structure of ${}^*\mathbb{R}$ and transfers by Łoś's theorem. Limit ordinal structure is set-theoretic and is not preserved by the ultrapower construction: ${}^*\mathbb{N}$ contains only successor-like infinite elements, not limit-like ones. The two results — transfer of density, failure of limit-ordinal structure — come from different theorems.

Section IV: Cantor Normal Form Bridge

Every ordinal below ϵ_0 has a unique Cantor normal form — a finite sum $a_1 \cdot \omega^{e_1} + a_2 \cdot \omega^{e_2} + \dots$ with strictly decreasing exponents $e_1 > e_2 > \dots$ and each coefficient a nonzero natural number (a nonzero ordinal strictly below ω). In Lean: `NONote (Mathlib.SetTheory.Ordinal.Notation)`. The encoding `cnfToZp2` maps each such ordinal to $\mathbb{Z}_2$ via structural recursion on the Cantor normal form.

Definition: `cnfToZp2` (Gentzen.lean §IV)

`cnfToZp2 : NONote →  $\mathbb{Z}_2$`

Base: `cnfToZp2(0) = 0`

Recursive: `cnfToZp2( $\omega^e \cdot n + a$ ) =  $2^{(v_2(\text{cnfToZp2}(e)) + 1)} \cdot n + \text{cnfToZp2}(a)$  [ $n : \mathbb{N}^+$ ]`

where v_2 denotes the 2-adic valuation.

Valuation of the n -th tower stage:

`cnfToZp2(towerNONote 0) = 0` (Lean: `PadicInt.valuation 0 = 0`; standard: $v_2(0) = +\infty$)

`cnfToZp2(towerNONote 1) = 2` (valuation 1)

`cnfToZp2(towerNONote 2) = 4` (valuation 2)

`cnfToZp2(towerNONote n) =  $2^n$` (valuation n)

Theorem: <code>cnfToZp2_tower_valuation</code>
$\forall n : \mathbb{N}, (\text{cnfToZp2 } (\text{towerNONote } n)).\text{valuation} = n$
The 2-adic valuation of the n -th tower stage encoding equals n .
Proof: by induction; successor step uses <code>PadicInt.valuation_pow</code> .
Lean purity: <code>[propext, Classical.choice, Quot.sound]</code> . ✓

Theorem: <code>cnfToZp2_valuation_unbounded</code>
$\forall k : \mathbb{N}, \exists \alpha : \text{NONote}, k \leq (\text{cnfToZp2 } \alpha).\text{valuation}$
The 2-adic valuation is unbounded across ordinals below ε_0 .
Proof: <code>towerNONote k</code> witnesses the bound.
Lean purity: <code>[propext, Classical.choice, Quot.sound]</code> . ✓

Theorem: <code>tower_converges_to_zero</code>
<code>Filter.Tendsto (fun n ↦ cnfToZp2 (towerNONote n)) Filter.atTop (nhds 0)</code>
The tower encodings converge to $0 = \perp$ in $\mathbb{Z}_2$.
Proof: each stage <code>cnfToZp2(towerNONote (n+1)) = 2ⁿ⁺¹</code> in $\mathbb{Z}_2$, so its 2-adic norm is $\ 2\ ^{n+1} = (1/2)^{n+1} \rightarrow 0$. Uses <code>Metric.tendsto_atTop</code> and <code>exists_pow_lt_of_lt_one</code> .
Lean purity: <code>[propext, Classical.choice, Quot.sound]</code> . ✓

Section V: Ordinal Tower Limit and Snap Threshold

This section connects the ordinal tower to the ZPE machine-phase snap. The cofinality theorem shows that the fundamental sequence approaches ε_0 from below. The lower-bound theorem shows that any monotone tower-aligned map must assign c_0 to all pre- ε_0 ordinals. A canonical witness snapping exactly at ε_0 is exhibited.

What this does NOT claim (§V)
That ε_0 is the proof-theoretic ordinal of PA (not claimed — see Remark R-L.1).
Any statement about formal provability in PA.
That ε_0 is the UNIQUE minimal snap boundary: <code>snap_threshold_is_epsilon_zero</code> shows no ordinal below ε_0 works (for maps satisfying the stated hypotheses), but does not rule out maps that snap at some ordinal strictly above ε_0 .
That the snap threshold result applies to all maps <code>Ordinal → MachinePhase</code> regardless of the monotonicity and tower-alignment hypotheses.

Theorem: fundamentalSeq_cofinal

$\forall \alpha : \text{Ordinal}, \alpha < \varepsilon_0 \rightarrow \exists n : \mathbb{N}, \alpha < \text{fundamentalSeq } n$

The fundamental sequence is cofinal in ε_0 : for any ordinal below ε_0 , some tower stage exceeds it.

Proof: $\varepsilon_0 = \text{nfp } (\omega^\cdot) 0$ (`epsilonZero_eq_nfp`), and `lt_nfp_iff` gives $a < \text{nfp } f \ b \leftrightarrow \exists n, a < f^n[n] \ b$.

Lean purity: [`propext`, `Classical.choice`, `Quot.sound`]. ✓

Theorem: snap_threshold_is_epsilon_zero

For any $\phi : \text{Ordinal} \rightarrow \text{MachinePhase}$ satisfying:

(a) `hmono`: $\forall \alpha \leq \beta, \text{join } (\phi \ \alpha) (\phi \ \beta) = \phi \ \beta$ (order-non-decreasing in the `ZPSemilattice` sense: $\phi \ \alpha \leq \phi \ \beta$)

(b) `h0`: $\forall n : \mathbb{N}, \phi \ (\text{fundamentalSeq } n) = c_0$

we have: $\forall \alpha < \varepsilon_0, \phi \ \alpha = c_0$

This is a lower bound: no ordinal below ε_0 is a snap point for maps satisfying (a) and (b). "Minimal" not "unique."

Proof: cofinality gives a stage above α ; monotonicity chains $\phi \ \alpha \leq \phi(\text{stage}) = c_0$; $c_0 = \perp$ forces $\phi \ \alpha = c_0$.

Lean purity: [`propext`, `Classical.choice`, `Quot.sound`]. ✓

Theorem: c1_epsilon_zero_identification

$\exists \phi : \text{Ordinal} \rightarrow \text{MachinePhase},$

$(\forall n : \mathbb{N}, \phi \ (\text{fundamentalSeq } n) = c_0) \wedge \phi \ \varepsilon_0 = c_1$

Witness: $\phi \ \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$.

One specific map sends all tower stages to c_0 and ε_0 to c_1 .

The stronger structural claim — an order-preserving morphism $\text{Ordinal} \rightarrow_o \text{MachinePhase}$ compatible with the $\text{CNF} \rightarrow \mathbb{Z}_2$ encoding — remains outside Lean scope: no type bridge between `Ordinal` and `MachinePhase` is defined in this library.

Lean purity: [`propext`, `Classical.choice`, `Quot.sound`]. ✓

Section VI: Kleene-Ordinal Fixed-Point Bridge

The ordinal fixed-point structure ($\varepsilon_0 = \text{nfp } (\omega^\cdot) 0$, $\omega^\cdot \varepsilon_0 = \varepsilon_0$) and the computational fixed-point structure (Kleene's recursion theorem, `roger_fixed_point_stability`) both carry `Classical.choice` in the proofs recorded here — parallel structure, not a proved isomorphism. The hypothesis `hfp` encodes that ordinal fixed points of $\omega^\cdot$ map to the snap state c_1 . Under this hypothesis plus monotonicity and tower alignment, ε_0 is the minimal snap threshold.

Theorem: snap_exactly_at_epsilon_zero

For any $\phi : \text{Ordinal} \rightarrow \text{MachinePhase}$ satisfying:

(a) hmono: order-non-decreasing ($\text{join } (\phi \alpha) (\phi \beta) = \phi \beta$ for $\alpha \leq \beta$)

(b) h0: every tower stage maps to c_0

(c) hfp: every fixed point of $\omega^\cdot$ maps to c_1

we have: $\phi \varepsilon_0 = c_1$ AND $\forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$

ε_0 is the minimal snap threshold: ϕ assigns c_1 first at ε_0 .

Note: "minimal" not "unique." A map satisfying these hypotheses could also assign c_1 to ordinals above ε_0 ; what is ruled out is any snap strictly before ε_0 .

Proof: upper bound from hfp + epsilonZero_fixedPoint; lower bound from snap_threshold_is_epsilon_zero.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: kleene_ordinal_snap_bridge

$\exists \phi : \text{Ordinal} \rightarrow \text{MachinePhase},$

$(\forall n : \mathbb{N}, \phi (\text{fundamentalSeq } n) = c_0) \wedge$

$(\forall \alpha, \omega^\alpha \alpha = \alpha \rightarrow \phi \alpha = c_1) \wedge$

$\phi \varepsilon_0 = c_1 \wedge$

$\forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$

Witness: $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$.

Note: the theorem name refers to the informal §VI conceptual parallel between Kleene recursion fixed points and ordinal fixed points of $\omega^\cdot$. The theorem itself is purely ordinal — no Code or eval appears.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Section VII: Canonical Snap Map — Full Closure

The canonical threshold map $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$ satisfies all three conditions of snap_exactly_at_epsilon_zero (hmono, h0, hfp). For this specific map, the snap identification is unconditional: all five conditions are simultaneously verified without free hypotheses.

Theorem: snap_map_mono

$\forall \alpha \beta : \text{Ordinal}, \alpha \leq \beta \rightarrow$

$\text{join } (\text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1)$

Theorem: snap_map_mono

(if $\beta < \varepsilon_0$ then c_0 else c_1) =

if $\beta < \varepsilon_0$ then c_0 else c_1

The canonical map is order-non-decreasing.

Proof: four cases (α/β vs ε_0). The case $\alpha \geq \varepsilon_0$ with $\beta < \varepsilon_0$ is impossible by $\alpha \leq \beta$. Each live case closes by rfl from the MachinePhase join definition.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: epsilon_zero_snap_canonical

$\exists \phi : \text{Ordinal} \rightarrow \text{MachinePhase},$

$(\forall \alpha \beta, \alpha \leq \beta \rightarrow \text{join} (\phi \alpha) (\phi \beta) = \phi \beta) \wedge [\text{hmono}]$

$(\forall n : \mathbb{N}, \phi (\text{fundamentalSeq } n) = c_0) \wedge [\text{h0}]$

$(\forall \alpha, \omega^\alpha = \alpha \rightarrow \phi \alpha = c_1) \wedge [\text{hfp}]$

$\phi \varepsilon_0 = c_1 \wedge [\text{snap}]$

$\forall \alpha, \phi \alpha = c_1 \rightarrow \varepsilon_0 \leq \alpha$ [minimality]

All five conditions verified for the explicit witness $\phi \alpha = \text{if } \alpha < \varepsilon_0 \text{ then } c_0 \text{ else } c_1$, with no free hypotheses.

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Theorem: snap_zp2_correspondence

Four independent facts about the same tower sequence:

(i) $\forall n : \mathbb{N}, \text{fundamentalSeq } n < \varepsilon_0$

(ii) $\forall n : \mathbb{N}, \phi (\text{fundamentalSeq } n) = c_0$ where ϕ is the canonical map

(iii) $\text{Filter.Tendsto} (\text{fun } n \mapsto \text{cnfToZp2} (\text{towerNONote } n)) \text{Filter.atTop} (\text{nhds } 0)$

(iv) $\phi \varepsilon_0 = c_1$

The same tower sequence witnesses both the ordinal approach to ε_0 and the 2-adic approach to 0. The full structural identification ($\varepsilon_0 \leftrightarrow \perp$ via a type bridge) remains outside Lean scope — see §V.

Proof: $\langle \text{epsilonZero_tower_lt}, \text{fun } n \mapsto \text{if_pos } \dots, \text{tower_converges_to_zero}, \text{if_neg } (\text{lt_irrefl } \varepsilon_0) \rangle$

Lean purity: [propext, Classical.choice, Quot.sound]. ✓

Remaining Formal Gap

The Remaining Gap

The identification of ZPE's MachinePhase element c_1 with the ordinal ϵ_0 via a formal type bridge remains outside Lean scope.

What is proved: a canonical map $\text{Ordinal} \rightarrow \text{MachinePhase}$ assigns c_1 exactly at ϵ_0 and nowhere earlier. What is not proved: a canonical ZPSemilattice morphism $\text{MachinePhase} \rightarrow \mathbb{Z}_2$ that would connect ZPE's $\perp = c_0$ to ZPB's $\perp = 0$ formally.

Two of the three pieces such a bridge was expected to need are built, in ZPM (ZeroParadox/Ordinal/Incompleteness.lean §I–§II), and the third is the one still open:

(1) $\text{snapEmbed} : \text{MachinePhase} \rightarrow \mathbb{Z}_2$ exists, sending $c_0 \mapsto 1$ and $c_1 \mapsto 0$ — the snap state to 2-adic zero, the pre-snap state to a unit (snapEmbed_c0 , snapEmbed_c1 , both by rfl)

(2) $\text{snapEmbed_mul_morphism}$ proves $\text{snapEmbed}(\text{join } a \text{ } b) = \text{snapEmbed } a \times \text{snapEmbed } b$, carrying join to multiplication because c_1 absorbs joins as 0 absorbs products. That is an absorbing-element morphism, not a ZPSemilattice morphism: $\mathbb{Z}_2$ carries no lattice $\perp$

(3) still open — no theorem derives the alignment from $\text{tower_converges_to_zero}$ through snapEmbed

So hfp is derived rather than assumed: $\text{hfp_from_epsilon_zero}$ obtains it from monotonicity together with $\text{h}\epsilon_0 : \phi \epsilon_0 = c_1$, and $\text{snap_unconditional}$ uses it in place of the free hypothesis of $\text{snap_exactly_at_epsilon_zero}$. What remains open is $\text{h}\epsilon_0$ itself — deriving $\phi \epsilon_0 = c_1$ from the 2-adic structure rather than taking it as an alignment hypothesis, which ZPM §II names the Classical.choice inversion conjecture. The canonical witness ($\text{epsilon_zero_snap_canonical}$) satisfies all five conditions without any such bridge; what the bridge would close is the gap between the two formal instances of $\perp$ across ZPE and ZPB.

Theorem Summary

Theorem	Section	Status
roger_fixed_point_stability	§II	Proved ✓
epsilonZero_fixedPoint	§III	Proved ✓
epsilonZero_eq_nfp	§III	Proved ✓
epsilonZero_eq_iSup	§III	Proved ✓
epsilonZero_tower_lt	§III	Proved ✓
epsilonZero_le_fixedPoint	§III	Proved ✓
fundamentalSeq_strictMono	§III	Proved ✓
tower_stage_zero / _one / _two	§III	Proved ✓
towerNONote_repr	§IV	Proved ✓
cnfToZp2_tower_valuation	§IV	Proved ✓

Theorem	Section	Status
cnfToZp2_valuation_unbounded	§IV	Proved ✓
fundamentalSeq_zp2_converges	§IV	Proved ✓
tower_converges_to_zero	§IV	Proved ✓
zpe_snap_ordinal_correspondence	§V	Proved ✓
epsilonZero_tower_bound	§V	Proved ✓
c1_epsilon_zero_identification	§V	Proved ✓
fundamentalSeq_cofinal	§V	Proved ✓
snap_threshold_is_epsilon_zero	§V	Proved ✓
snap_exactly_at_epsilon_zero	§VI	Proved ✓
kleene_ordinal_snap_bridge	§VI	Proved ✓
snap_map_mono	§VII	Proved ✓
epsilon_zero_snap_canonical	§VII	Proved ✓
snap_zp2_correspondence	§VII	Proved ✓

Axiom Purity

Every theorem in the Theorem Summary above carries axiom footprint: [propext, Classical.choice, Quot.sound]. Section I's table and ZP-K Section IV record where the ZP layers differ — Section I's own ZPJ/K row names bot_self_mem, which measures no axioms.

These are standard Mathlib infrastructure axioms (ordinal theory, p-adic analysis, computability). They are not ZP-L commitments.

The footprints are not uniform; ZP-K Section IV tabulates the measurements. In the computability layer Classical.choice is reached through Mathlib's numbering of program codes (Denumerable Code): it is present even in a computable instance with a constant code. ZP-K's machinePhaseKleene also picks botCode with Classical.choose, which is what makes that instance noncomputable. Whether the numbering's footprint is essential is not measured. Its presence is expected and documented.

Zero sorry in Gentzen.lean. Verified: lake build, May 2026.